

Configuration & Troubleshooting

1. Cisco CLI Modes

Cisco devices have different modes for security and control.

Mode	Prompt	Purpose	Command
User Mode	Router>	Basic view only	Default
Privileged Mode	Router#	Admin access	enable
Global Config	(config)#	Change device settings	configure terminal

Example

```
Router> enable
Router# configure terminal
Router(config)#
```

Simple meaning

- > → normal user
- # → admin
- (config)# → editing device settings

2. Basic Router Configuration

Step 1 — Enter Configuration Mode

```
Router> enable
Router# configure terminal
```

Step 2 — Set Router Name

```
Router(config)# hostname R1
```

Now router prompt becomes:

```
R1(config)#
```

Step 3 — Configure an Interface

Routers communicate through **interfaces (ports)**.

Example configuration:

```
R1(config)# interface gigabitEthernet0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
```

Explanation:

Command	Meaning
interface g0/0	Select the port
ip address	Assign IP address
no shutdown	Turn the port ON

Important:

Cisco interfaces are **off by default**.

Without `no shutdown` the interface will stay **disabled**.

3. Basic Switch Configuration

Switches work automatically but need an **IP for management**.

```
Switch> enable
Switch# configure terminal
```

```
Switch(config)# hostname S1
```

Assign Management IP

```
S1(config)# interface vlan 1  
S1(config-if)# ip address 192.168.1.2 255.255.255.0  
S1(config-if)# no shutdown
```

Set Default Gateway

```
S1(config)# ip default-gateway 192.168.1.1
```

Explanation:

The switch will send traffic for other networks to the router.

4. Static Routing

Routers only know **directly connected networks**.

To reach another network, we must create a **static route**.

Syntax

```
ip route DESTINATION_NETWORK SUBNET_MASK NEXT_HOP
```

Example

```
R1(config)# ip route 10.0.0.0 255.255.255.0 192.168.1.2
```

Meaning:

To reach network 10.0.0.0

```
send packets to router 192.168.1.2
```

5. Default Route

Default route is used for **unknown networks (usually internet)**.

```
R1(config)# ip route 0.0.0.0 0.0.0.0 192.168.1.1
```

Meaning:

```
If router doesn't know where to send traffic  
send it to 192.168.1.1
```

Usually this is the **ISP router**.

6. Important Troubleshooting Commands

These commands help find network problems.

Check Interface Status

```
show ip interface brief
```

Common results:

Status	Meaning
up/up	Working
administratively down	Interface disabled
down/down	Cable problem

Check Routing Table

```
show ip route
```

Important codes:

Code	Meaning
C	Connected
S	Static route
O	OSPF route

Check Connectivity

Ping

```
ping 192.168.1.1
```

Meaning:

```
"Are you reachable?"
```

Traceroute

```
traceroute 8.8.8.8
```

Shows the **path packets take to reach the destination**.

7. Common Network Problems

Problem	Cause	Fix
Ping fails	Interface disabled	no shutdown
Ping fails	Wrong IP address	Check device IP
Cannot reach internet	Missing gateway	Configure default gateway
Cannot reach other network	Missing route	Add static route

8. Save Configuration

If you reboot without saving, configuration is lost.

Save it using:

```
copy running-config startup-config
```

OR

```
write memory
```

Meaning:

```
running-config → current settings
startup-config → saved settings used after reboot
```

Quick Command Summary

```
enable
configure terminal
hostname
interface
ip address
no shutdown
ip route
show ip interface brief
```

```
show ip route  
ping  
traceroute  
copy running-config startup-config
```

One Line Summary

Router configuration = Assign IP + Enable interface + Add routes

Troubleshooting = Use show commands + ping + traceroute